Raspberry Pi Foundation

Lesson 3: Classification

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If you are an educator, ask your students to complete a short survey: rpf.io/exai-st.

Learners will explore how machine learning (ML) models are created with supervised learning. They will build their understanding of how labelled training data is used to train classification models by interacting with Quick, Draw!, an interactive AI tool. Finally, they will examine how confidence scores are used in ML model predictions.

Objectives

Objective 1	Describe how supervised learning is used to create a classification model using classes and labelled data
Objective 2	Explain how confidence scores are used in classification predictions

Keywords

- Supervised learning
- Classification
- Class
- Label
- Confidence score

Lesson outline

Activity	Objective	Description	Time
Activity 1: Rule-based or data-driven?		Learners identify whether different systems are rule-based or data-driven.	5 mins
Activity 2: Different types of machine learning	1	Learners explore different types of ML. They are told how a classification model is trained with supervised learning using labelled data. They complete a practical activity with a classification tool.	20 mins
Activity 3: Classification and computer vision	2	Learners explore how classification is used in computer vision and learn about confidence scores in ML predictions.	15 mins
Optional: Classification in the real world	1	Learners think about uses of classification in the real world, including the challenges.	~15 mins
Exit ticket	1, 2	Learners review key concepts about classification by answering three short questions.	5 mins

Resources

Computers with internet access

Note: If you do not have access to computers for all learners, please see the alternative approach for Activity 2 described in the lesson walkthrough.

Links
- quickdraw.withgoogle.com
- quickdraw.withgoogle.com/data

Printed/Digitally distributed
- Worksheet 2: Quick, Draw!
- Worksheet (optional): Classification in the real world

Subject knowledge

Activity 1: You will need to be able to support learners in identifying whether the systems described are rule-based or data-driven. Answers are provided in the slide deck.

Activity 2: This activity introduces the three main different types of machine learning. It then focuses on supervised learning. It explores how supervised learning is used to train a classification model. Classification is where a machine learning model is used to classify (categorise) data into groups (called 'classes'). The model will produce a 'label' (a prediction of which class the data belongs to). We recommend that you review these key terms before the lesson.

In addition, we recommend that you familiarise yourself with the Quick, Draw! tool (quickdraw.withgoogle.com), including locating the dataset (quickdraw.withgoogle.com/data), to help you guide the learners as they complete the activity.

Activity 3: You will introduce how classification is used in **computer vision**, and one new concept, **confidence scores**. We recommend that you familiarise yourself with these concepts to help you answer any questions the learners may have. To help you discuss these concepts with learners, example questions and answers are provided in the lesson walkthrough below.

Assessment opportunities

Objective	Description
1	Activity 2: You can assess learners' ability to describe how supervised learning can be used to create classification models in the Quick, Draw! activity, where learners will discuss the importance of large datasets in creating ML models. Activity 3: You can assess learners' understanding of classes by assessing how confidently they can apply what they have learned to computer vision Exit ticket: You can assess learners' answers to the questions.
2	Exit ticket: You can assess learners' answers to the questions.

Lesson walkthrough

The **Success criteria** on slide 7 translate lesson objectives into actionable, student friendly targets. They clearly define what the students should be able to do by the end of the lesson.

Slide deck labels appear at the top of each slide to help you navigate the activities in the lesson.

This lesson includes an **optional activity** that will take around 15 minutes. If you do not want to include it, please skip slides 27–29. These slides have been hidden by default.

5 mins	**Activity 1: Rule-based or data-driven?**
Slides 2–6	Ask learners to identify whether each example system (a traffic light, an app that recommends movies, and a chatbot) is data-driven or rule-based.
	Educator guidance Ask learners to give a thumbs up if they think a system is data-driven or a thumbs down if they think it is rule-based, so that all learners give an answer. Before you reveal the answers using the slide animations, ask a few learners to explain why they think a system is data-driven or not. If you feel that the words 'recommends' and 'generate' are too complex for your learners, you could change 'recommends' to 'suggests' and 'generate' to 'produce' on the slides. We have used the word 'recommends' as the example refers to a recommendation algorithm, and 'generate' as the example refers to a generative AI application — a chatbot.
20 mins	**Activity 2: Different types of machine learning** **Resource required: Worksheet 2: Quick, Draw!**
Slide 8 and 9	Explain to your learners that there are three main types of machine learning. Use the animations and the descriptions on the slide to highlight the key features and an example of each type. Slide 9 highlights supervised learning as this is the type that the lessons will focus on from this point.
Slide 10	Use this slide to remind your learners how machine learning is used to train a model. Training will identify patterns that are used by the model to make predictions. Highlight that supervised learning uses data labelled by humans
Slide 11	In this slide, box one is highlighted to emphasise that we will now look at how the data is labelled.
Slide 12	The slide explains that supervised learning can be used to create a classification model. The slide goes on to explain a classification model is used to categorise data and gives three animated examples.
Slide 13	Highlight that a classification model is trained using a large set of data that has been labelled by humans, and that humans group this data into classes. The images on the slide show examples of two classes: 'Apple' and 'Orange'. These contain multiple images of apples and oranges that have each been labelled with the class they belong to, 'Apple' or 'Orange'.
Slide 14	Box two is now highlighted to demonstrate that the lesson will focus on how patterns are identified in the labelled training data

Slide 15	Explain that a classification model is trained with example data (training data). This training data has already been labelled by a human. Highlight that the more training data you use for each class, the more accurate the model's predictions will be. **Educator guidance** Note that this rule depends on the quality of the data used for the training. If the training data does not accurately reflect what is being modelled — for example, if an image of a dog is included in the training data for a model designed to identify cats — then this will not improve the accuracy of the model.
Slide 16	Box three is now highlighted to demonstrate that the lesson will focus on the predictions made using new data
Slide 17	Show what happens once training is complete: New data can be provided to the model and it will predict a label from the options it has been trained with. You can mention the confidence score that comes with the prediction, to introduce the idea of uncertainty.
Slide 18	Instruct learners to go to quickdraw.withgoogle.com and play one round. Inform learners that their drawings may be added to an open-source dataset used to train AI models. **Educator guidance** Quick, Draw! will instruct learners to draw a specific object within 20 seconds. The ML model will then attempt to classify what the learners are drawing in real time. Each round consists of six drawings, and learners should complete at least one full round. This will allow them to see how the model produces predictions and how its accuracy changes based on the quality of the input data (their drawings). **Alternative approach** If you do not have access to computers for all learners, learners can complete the activity in pairs or larger groups, or you can demonstrate the activity by presenting your screen. You could also ask 2–3 students to try the activity on your computer.
Slide 19	Hand out **Worksheet 2: Quick, Draw!** and explain to the learners that the model's ability to classify their drawings is based on a huge dataset of millions of labelled drawings contributed by people online. Direct them to the dataset, available at quickdraw.withgoogle.com/data. This is where they can explore the raw data that was used to train the model. Instruct them to explore the different classes within the dataset, then choose a class to look at and answer the questions in the first part of their **worksheets**. **Educator guidance** The questions will help you highlight some important concepts: • **How many drawings does the class include?** This demonstrates the sheer volume of data used for training. • **Why is it important to have a large dataset for each class?** This helps start a discussion about the importance of the quantity of data in machine learning: the more relevant example data is used when training a classification model, the more accurate the model will be. **Alternative approach**

	If you are unable to print the worksheet, use slide 14 to show the questions, and ask learners to write their answers on a sheet of paper instead.
Slide 20	Revisit the concepts of 'label' and 'class', using an example from the Quick, Draw! dataset ('bird') to solidify these definitions. Emphasise that the ML model gives a drawing a 'label', which is a prediction of what the drawing shows, while the 'class' is the category that drawing belongs to.
15 mins	**Activity 3: Other types of classification**
Slide 21	Explain that learners will now explore how classification is used within computer vision. Introduce computer vision as when a model is used to identify an object or range of objects in a photo or video. Highlight that models can detect many objects in a single image. Models are trained with images that might contain individual objects or multiple objects. Explain why computer vision models are helpful by highlighting that these models can detect multiple objects in a photo or video, which is useful for technologies such as driverless cars Your learners are shown a collection of training data that will be used to train the classification model for the computer vision system. Ask your learners what classes this data could be grouped into?
	Educator guidance You could ask learners what other classes of objects a driverless car might need to detect, for example: • Traffic lights • Curbs • Other vehicles • Signs
Slide 22	This slide reveals the answer: car, dog, human. Ask your learners what is identified within these classes during training?
Slide 23	This slide reveals the answer: patterns Ask your learners to answer the question on the board by showing one or two fingers. The answer is predictions as classification models will not output definitive answers.
Slide 24	Explain that a classification model predicts a label and also a confidence score.
Slide 25	Explain that confidence scores indicate how likely it is that a prediction is accurate.
	Educator guidance Use the image on the slide to emphasise how a classification model classifies new data and produces predictions. Highlight that a classification model is trained on large quantities of image data, with classes like 'Car', 'Dog', and 'Human'. When new data is presented, the model compares patterns identified in the new data with patterns identified in the training data to predict the class that the new data belongs to.

Slide 26 — Use this slide to highlight what a high confidence score means.

15 mins

Optional activity: Classification in the real world

Resource required: Worksheet: Classification in the real world

Educator guidance

This is an optional extension activity. If you do not want to include it, please skip slides 25–27.

Slide 27 **Optional** — Explain that in this activity, learners will work in teams to explore how classification is used in the real world.

Slide 28 **Optional** — Divide your class into groups of 3–5 learners. Talk through the slide and instruct learners to use **Worksheet: Classification in the real world** or a sheet of paper to answer the questions on the slide.

Educator guidance

You could use **Worksheet: Classification in the real world** to collect your learners' answers. If you are unable to print the worksheet, use slide 26 to show the questions and ask learners to write their answers on a sheet of paper instead.

Walk around the class to answer questions and provide guidance to your learners.

Below are some example answers.

Text tasks

News article categorisation:

What classes could you use?	The model will classify different types of news articles. We will create a class for each type, for example, 'sports', 'politics', and 'entertainment'.
How will you collect the data to train your model?	We will collect different types of news articles from different newspapers.
What problems might you have?	We may need permission from the news publisher or other rights holders, as the news articles may be copyrighted, may be protected by press publisher rights, or may contain personally identifiable information, such as the names of authors. The model might apply an incorrect label to a news article that contains slang or sarcasm that does not appear in the training data.

Social media comment classification:

What classes could you use?	The model will classify comments that contain

	harmful content and comments that do not. We will create the classes ‘harmful’ and ‘not harmful’.
How will you collect the data to train your model?	We will collect existing comments from a social media platform.
What problems might you have?	We may need permission from the platform (on behalf of its users) to use user comments to train the model. Even if the platform can give us permission, we should consider whether it is fair to the users to use their comments as training data, and any rights they as individuals may have in their comments. The model may apply incorrect labels to comments that contain emojis, slang, or abbreviations that do not appear in the training data.

Numbers task

Heart rate classification in fitness trackers:

What classes could you use?	The model will classify different heart rates. We will create the classes ‘resting’, ‘target’, and ‘abnormal’.
How will you collect the data to train your model?	We will collect medical data that demonstrates these different heart rates.
What problems might you have?	We will need to ask patients to give consent for us to use their data to train the model, and we will need to ensure our use of the data complies with regulation in this space (including around use of patient data and using AI in healthcare). We will need to make users aware that the predictions may sometimes be inaccurate. The model might sometimes produce incorrect predictions due to variations across individuals, missing data (for example, when users do not wear the tracker), and outliers.

Audio tasks

Identifying different musical instruments in a song:

What classes could you use?	The model will classify different musical instruments. We will create classes for each of

	these instruments, such as ‘guitar’, ‘violin’, ‘drums’, and ‘piano’.
How will you collect the data to train your model?	We will collect music recordings with those instruments in.
What problems might you have?	We may need permission from the people who own the song to use it in this way, particularly if the song is copyrighted. The recordings will need to be good quality, without any background noise.

Classifying command words for a smart speaker:

What classes could you use?	The model will classify words to identify whether they are command words (such as ‘play’, ‘stop’, and ‘next’). We will create a class for each command word.
How will you collect the data to train your model?	We will record people with lots of different accents and voices.
What problems might you have?	It might be hard to record enough people with different accents, voices, and speaking styles, as well as enough people with a range of speaking difficulties, to make the system reliable for all users.

Slide 29 **Optional**	Allow each group to present for about 1 minute. Depending on time, encourage learners to ask each other questions.
5 mins	**Exit ticket**
Slide 30	Ask the ‘true or false’ question on the slide, then use the animation to reveal the answer. Emphasise that pre-labelled data is a core idea of supervised learning. Invite learners to reflect on earlier activities (like using Quick, Draw!) where labelled data was essential.
Slides 31–32	Use the multiple choice question on slide 29 to encourage learners to recall why confidence scores are important, then reveal the answer on slide 30. If necessary, use slide 23 to remind learners what confidence scores are.